

Integral Homology of the Multiplicity-11 Singleton Stratum

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Abstract

For the reduced cellular complex associated with the closed principal stratum of monic real polynomials generated by the singleton composition (11), we compute the integral homology in every admissible odd degree. It is $\mathbb{Z}$ in homological degree 1 when the polynomial degree is 11, and it vanishes in every degree for all odd polynomial degrees greater than 11. The proof uses the geometric realization of the cellular complex: the uncompactified stratum is homeomorphic to $\mathbb{R} \times \mathcal{Q}_{d-11}$, where $\mathcal{Q}_{d-11}$ is the space of nonnegative monic polynomials. At degree 11 its one-point compactification is S^1 ; in every larger admissible degree, adding an unbounded nonnegative constant to the second factor contracts the one-point compactification to infinity. This proves the $n = 11$ specialization of the odd case of Shapiro's one-part resonance conjecture.

1 Statement and context

A composition is a finite ordered tuple $\eta = (\eta_1, \dots, \eta_r)$ of positive integers. Write

$$|\eta| = \sum_{i=1}^r \eta_i, \quad \ell(\eta) = r, \quad |\eta|' = |\eta| - \ell(\eta).$$

Starting from a composition, one may merge two adjacent parts or insert a part equal to 2 in any gap. We write $\eta \preceq \omega$ when η can be obtained from ω by finitely many such operations, including no operation.

For an odd integer $d \geq 11$, let

$$C_*(11; d) = \mathbb{Z}\{\eta : \eta \preceq (11), |\eta| \leq d\}, \quad \deg_d(\eta) = d - |\eta|'.$$

If $M_i\eta$ merges parts i and $i + 1$, and $I_g\eta$ inserts 2 after the first g parts, the differential is

$$\partial\eta = \sum_{i=1}^{r-1} (-1)^{i-1} M_i\eta + \sum_{g=0}^r (-1)^g I_g\eta, \tag{1}$$

where insertion terms of norm greater than d are omitted and coincident terms are combined. Both operations raise $|\eta|'$ by one, so ∂ lowers the displayed degree by one.

This is the reduced cellular complex described by Shapiro [1, formulas (1)–(2)]. The same source identifies its homology with the reduced integral homology of the one-point compactification of the corresponding closed principal stratum. Its Conjecture 10 predicts the homology of every one-part composition and reports finite verification for $n \leq 10$ and $d \leq 40$ [1, Conjecture 10]. Here we prove exactly the multiplicity-11 specialization; the even one-part case and compositions with more than one part are outside the scope of the result.

Theorem 1. *For every odd integer $d \geq 11$ and every integer j ,*

$$H_j(C_*(11; d), \partial) \cong \begin{cases} \mathbb{Z}, & (d, j) = (11, 1), \\ 0, & \text{otherwise.} \end{cases}$$

In particular, the vanishing for $d > 11$ is integral and excludes torsion as well as free homology.

2 The closed singleton stratum

Identify monic real polynomials of degree d with their d nonleading coefficients, hence with $\mathbb{R}^d$. Write $d = 11 + 2k$ and set

$$\mathcal{Q}_{2k} = \{q \in \mathbb{R}[x] : q \text{ is monic of degree } 2k \text{ and } q(x) \geq 0 \text{ for every } x \in \mathbb{R}\}.$$

The geometric description in [1, formula (3)] specializes to the closed singleton stratum

$$X_{11,d} = \{(x - z)^{11}q(x) : z \in \mathbb{R}, q \in \mathcal{Q}_{2k}\}. \quad (2)$$

Let $X_{11,d}^+$ denote its one-point compactification. The cellular identification just recalled gives

$$H_j(C_*(11; d), \partial) \cong \tilde{H}_j(X_{11,d}^+; \mathbb{Z}). \quad (3)$$

Thus it remains to determine the topology of $X_{11,d}^+$.

Lemma 2. *The map*

$$\Phi : \mathbb{R} \times \mathcal{Q}_{2k} \longrightarrow X_{11,d}, \quad \Phi(z, q) = (x - z)^{11}q(x),$$

is a homeomorphism.

Proof. Surjectivity is (2), and continuity is immediate in coefficient coordinates. A nonzero nonnegative real polynomial has only even-multiplicity real zeros. Consequently $(x - z)^{11}q(x)$ changes sign at z and nowhere else. This remains true if $q(z) = 0$, because that zero adds an even multiplicity. Hence z is intrinsically the unique sign-changing real root; after recovering z , polynomial division recovers q . This proves injectivity.

We verify continuity of the inverse. Suppose

$$p_m = (x - z_m)^{11}q_m \longrightarrow p = (x - z)^{11}q$$

in coefficient space. Since these are monic polynomials and their coefficients converge, Cauchy's root bound bounds all their roots uniformly; in particular, (z_m) is bounded. From any subsequence choose a further subsequence with $z_m \rightarrow w$. Repeated synthetic division has quotient coefficients that depend continuously on the dividend and the proposed root. It follows along this subsequence that

$$p = (x - w)^{11}q_w, \quad q_m \longrightarrow q_w$$

for a monic polynomial q_w of degree $2k$. Coefficient convergence implies pointwise convergence, so $q_w(x) \geq 0$ for every real x . The uniqueness just proved forces $w = z$ and $q_w = q$. Every convergent subsequence of (z_m) therefore has the same limit z , whence $z_m \rightarrow z$; synthetic division then gives $q_m \rightarrow q$. Thus Φ^{-1} is continuous. $\square$

3 Contraction at infinity

For $k = 0$, the space $\mathcal{Q}_0$ consists only of the constant polynomial 1. Lemma 2 then gives $X_{11,11} \cong \mathbb{R}$, and therefore $X_{11,11}^+ \cong S^1$.

The positive-excess case has a different topology. The key point is that the cone of nonnegative polynomials has a one-sided constant direction.

Lemma 3. *If $k \geq 1$, the one-point compactification of $Y = \mathbb{R} \times \mathcal{Q}_{2k}$ is contractible.*

Proof. The set $\mathcal{Q}_{2k}$ is closed in its finite-dimensional coefficient space: it is the intersection, over $x \in \mathbb{R}$, of the closed conditions $q(x) \geq 0$. Hence Y is locally compact and Hausdorff, and its one-point compactification $Y^+ = Y \cup \{\infty\}$ is defined.

For $a \geq 0$, put

$$T_a(z, q) = (z, q + a).$$

Since $k \geq 1$, adding a constant preserves both the degree and monicity of q , and nonnegativity is preserved as well. Each T_a is a continuous proper self-map of Y . Define

$$H(y, t) = \begin{cases} T_{t/(1-t)}(y), & y \in Y, 0 \leq t < 1, \\ \infty, & y \in Y, t = 1, \\ \infty, & y = \infty, 0 \leq t \leq 1. \end{cases} \quad (4)$$

We check the continuity at the added point and at $t = 1$.

Let $K \subset Y$ be compact. The constant coefficients of the polynomial components occurring in K are bounded. If $q \in \mathcal{Q}_{2k}$, then $q(0) \geq 0$, so the constant coefficient of $q + t/(1-t)$ is at least $t/(1-t)$. It follows that, as $t \rightarrow 1$, the image of every point of Y leaves K uniformly. This proves continuity at all points with time coordinate 1, including $(\infty, 1)$.

Now fix $t_0 < 1$. On a sufficiently small closed interval about t_0 , the numbers $a(t) = t/(1-t)$ lie in some compact interval $[0, A]$. The union

$$\bigcup_{0 \leq a \leq A} T_a^{-1}(K)$$

is compact: membership in it bounds z and every polynomial coefficient, because the points $(z, q+a)$ lie in K and a is bounded; it is also closed in the finite-dimensional coefficient model. Its complement is therefore a neighborhood of infinity that all the relevant T_a map outside K . This proves continuity at (∞, t_0) . Thus (4) is a continuous homotopy on Y^+ , with $H(-, 0)$ the identity and $H(-, 1)$ the constant map to infinity. $\square$

Proof of Theorem 1. At $d = 11$, the preceding discussion gives $X_{11,11}^+ \cong S^1$, whose reduced integral homology is $\mathbb{Z}$ in degree 1 and zero otherwise. If $d > 11$ is odd, then $d = 11 + 2k$ with $k \geq 1$. Lemma 2 identifies $X_{11,d}$ with $Y = \mathbb{R} \times \mathcal{Q}_{2k}$, and a homeomorphism between locally compact Hausdorff spaces extends to their one-point compactifications. By Lemma 3, $X_{11,d}^+$ is contractible. The conclusion now follows from (3). $\square$

4 Exact finite corroboration

The proof above is uniform in d and does not infer the theorem from a finite scan. Exact integer calculations nevertheless provide checks on the cellular signs and grading. At $d = 13$, the only chain groups are

$$C_3 = \mathbb{Z}\{(11)\}, \quad C_2 = \mathbb{Z}\{(2, 11), (11, 2)\}, \quad C_1 = \mathbb{Z}\{(13)\}.$$

With the signs in (1),

$$\partial_3(11) = (2, 11) - (11, 2), \quad \partial_2(a(2, 11) + b(11, 2)) = (a + b)(13).$$

The product of the two boundary matrices is zero, both nonzero Smith invariants are 1, and the complex is integrally acyclic.

A Python implementation using exact SymPy integer arithmetic also checked $d = 11, 13, 15, 17$. Generator sets produced independently by the descendant operations and by the singleton membership criterion agreed in every case; $\partial^2 = 0$ held; and the Smith data gave $\mathbb{Z}$ in degree 1 at $d = 11$ and zero homology in the other three cases. These computations are corroborative only: integral acyclicity for all $d > 11$ follows from the contraction in Lemma 3.

As broad adjacent methodological context, Zeng [2] also pairs an exact polynomial calculation with a second symbolic certificate, there for a finite counterexample concerning preorder support polynomials. That work does not concern root-multiplicity strata or their homology and is not used in the proof above; the relationship is limited to the practice of making computer-assisted polynomial checks independently replayable.

Scope and assistance disclosure

The argument uses the cellular identification and closed-stratum formula in [1]. It proves only the multiplicity-11 assertion stated in Theorem 1; it does not settle the even one-part prediction or the conjectures for general compositions in that source.

OpenAI Codex assisted with proof exploration, adversarial checking, exact small-case computation, and preparation of the exposition. The exact checks used Python and SymPy 1.14.0, including two independent generator enumerations, verification of $\partial^2 = 0$, and Smith normal forms over the integers. No finite computation is used as a proof of the quantified statement; the theorem rests on the geometric argument above. The author reviewed the mathematical claims and is responsible for the final text.

References

- [1] Boris Shapiro. Real polynomials with given multiplicities of real roots: Complete conjectural description of homology, 2026.
- [2] Zijian Zeng. An eight-element counterexample to real-rootedness of preorder support polynomials. SSRN working paper 7385158, 2026.